Table of Contents

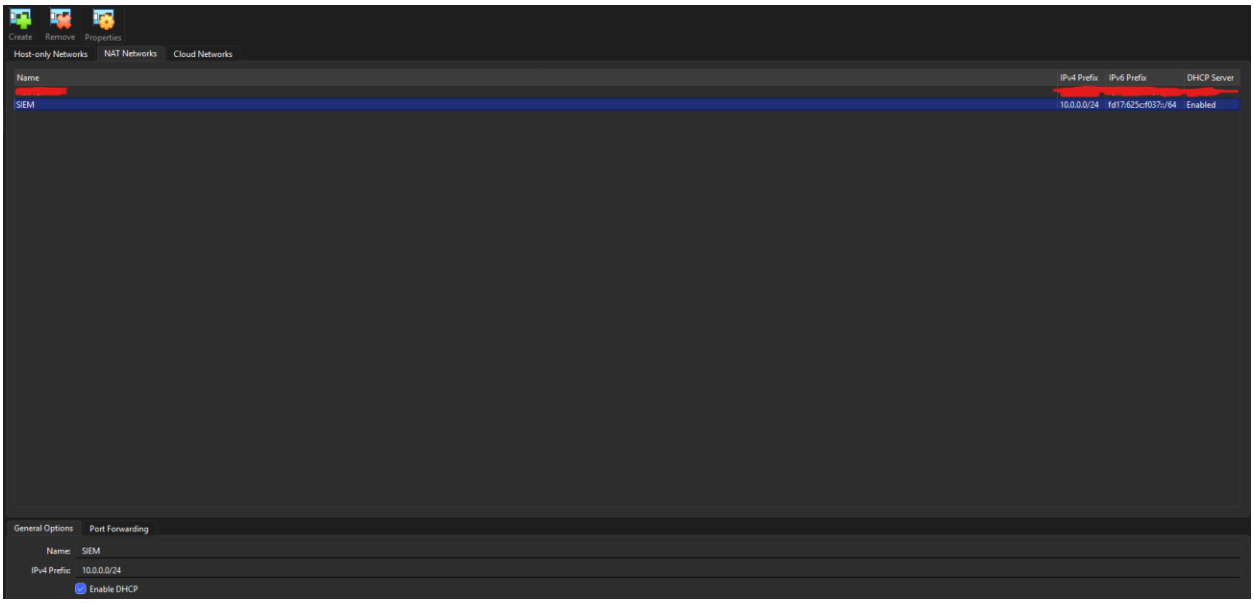
Oracle Box Virtual Machine Configurations 2
Nat Network 2
Ubuntu Splunk Server 2
Windows VM 3
Kali VM 3
Ubuntu VM 4
Setting up Splunk! 4
Splunk receiving port configuration 5
Splunk Index Creation 5
Splunk Forwarder Setup 7
Windows Setup 7
Kali Setup 10
Ubuntu Setup 11
Auto Activity Script Making 14
Windows Script 14
Ubuntu script 15
Kali script 16
Activity Result 17

Oracle Box Virtual Machine Configurations

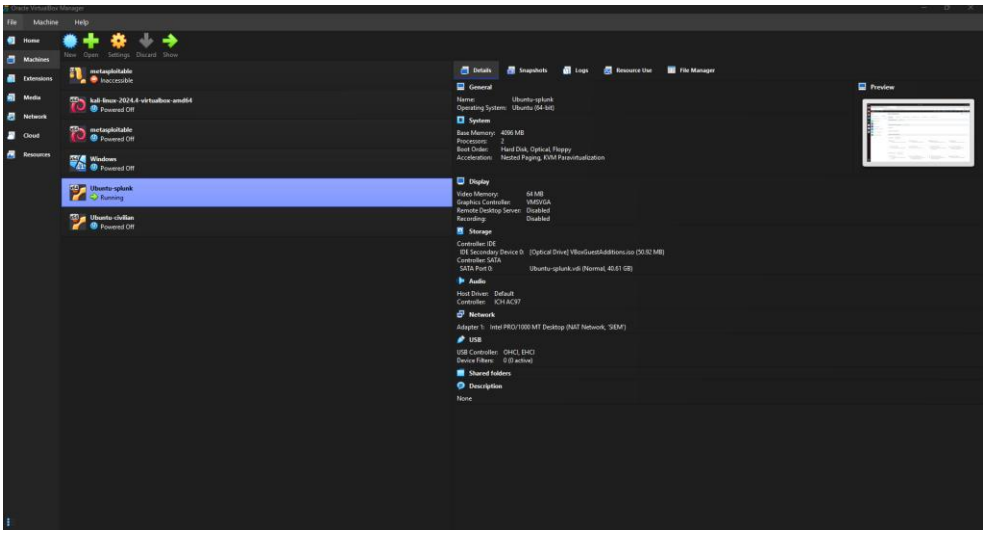
Oracle Virtual Box setup. The machines being used are the kali, ubuntu-splunk and ubuntu-civilian and windows machines.

Nat Network

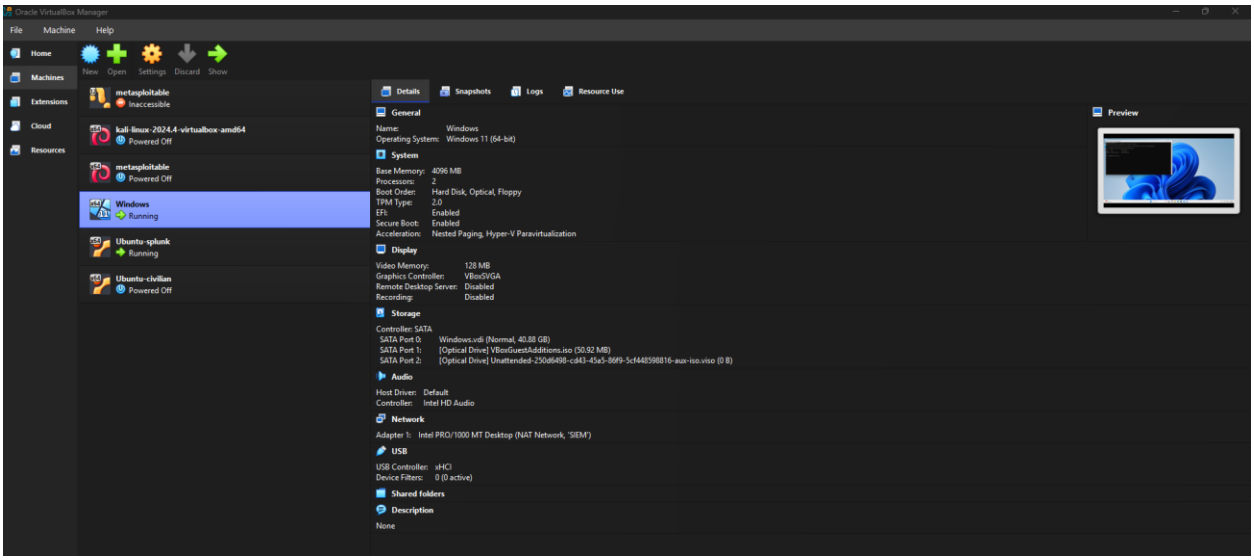
To communicate with each other I set up a NAT network first and named it SIEM



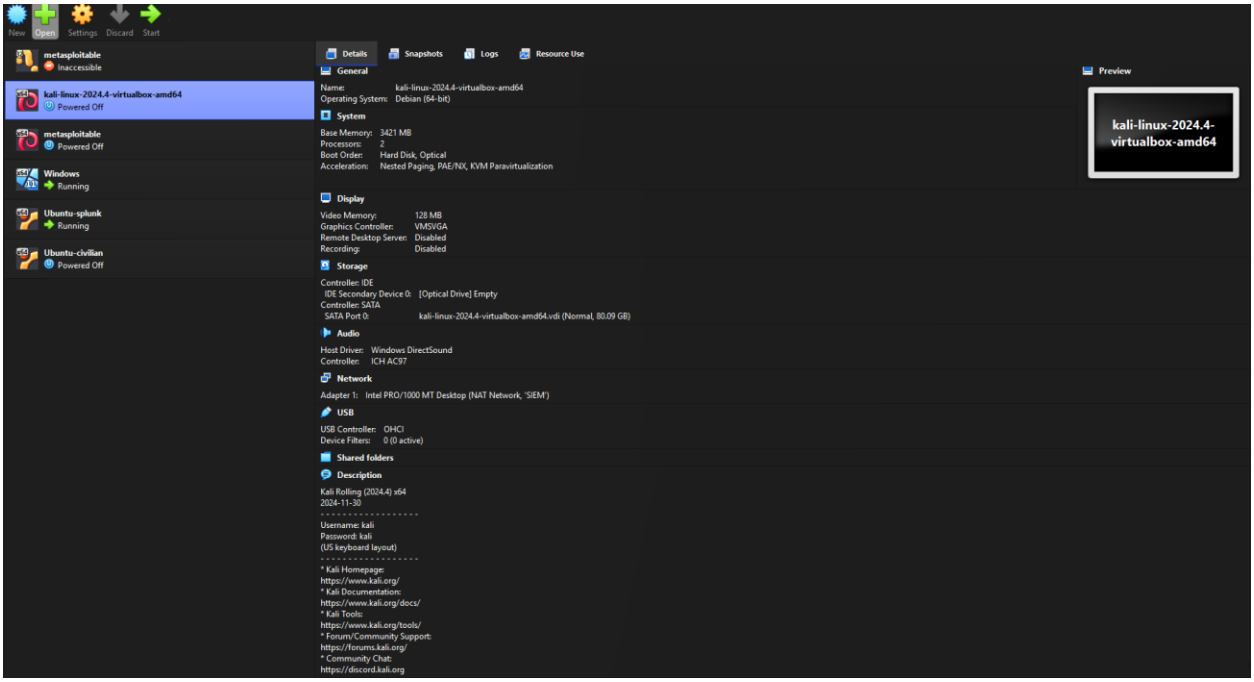
Ubuntu Splunk Server



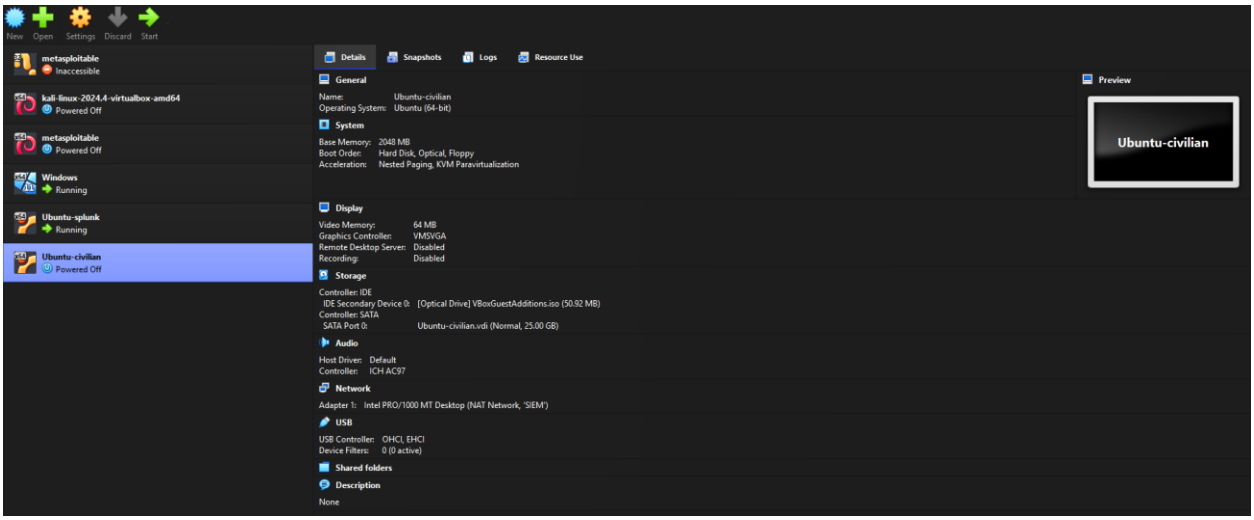
Windows VM



Kali VM

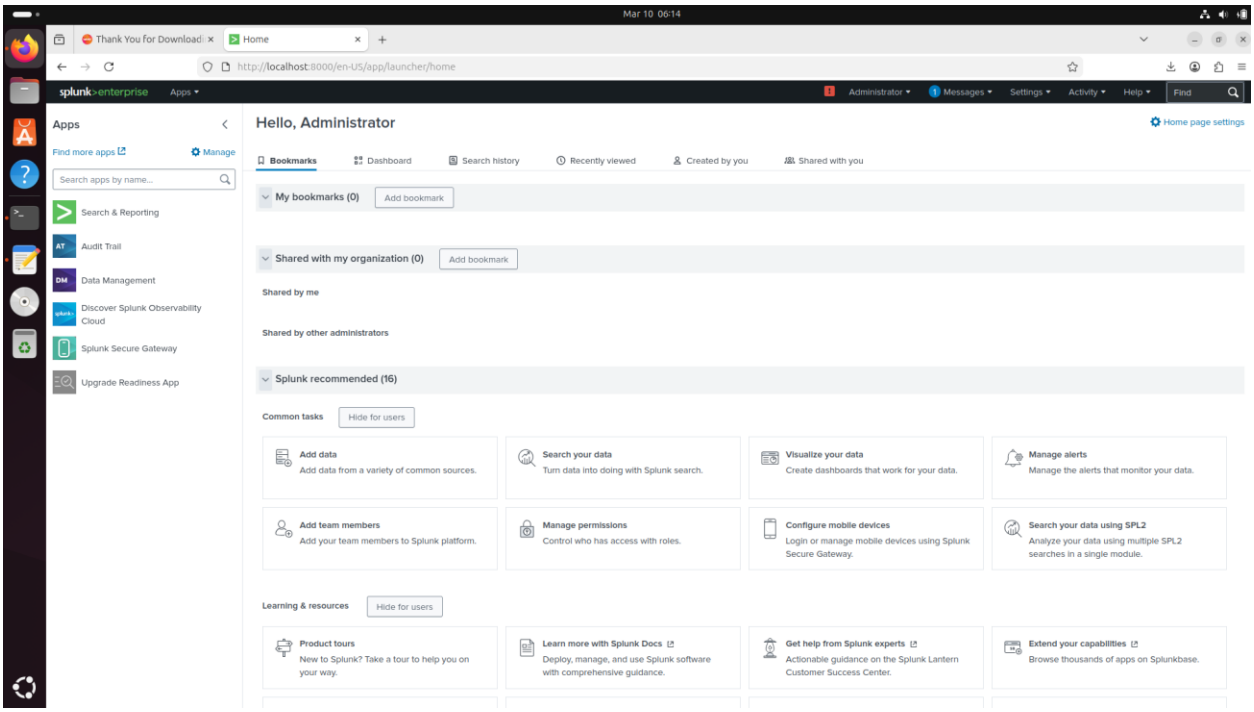


Ubuntu VM



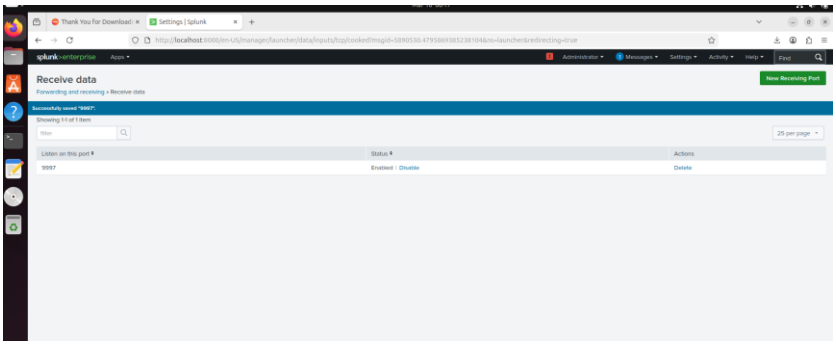
Setting up Splunk!

Splunk Dashboard after first login



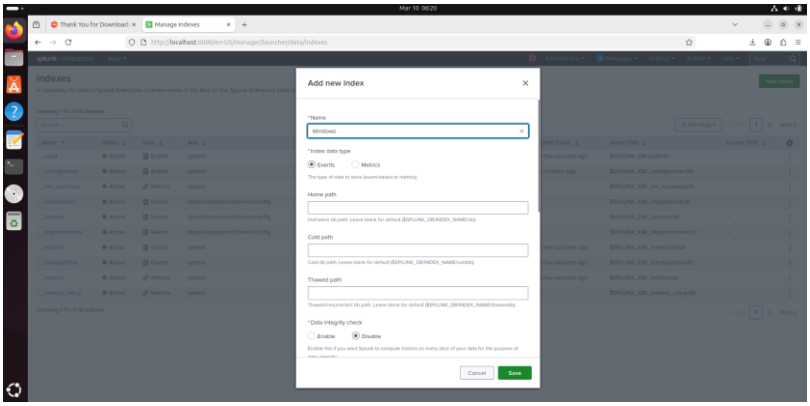
Splunk receiving port configuration

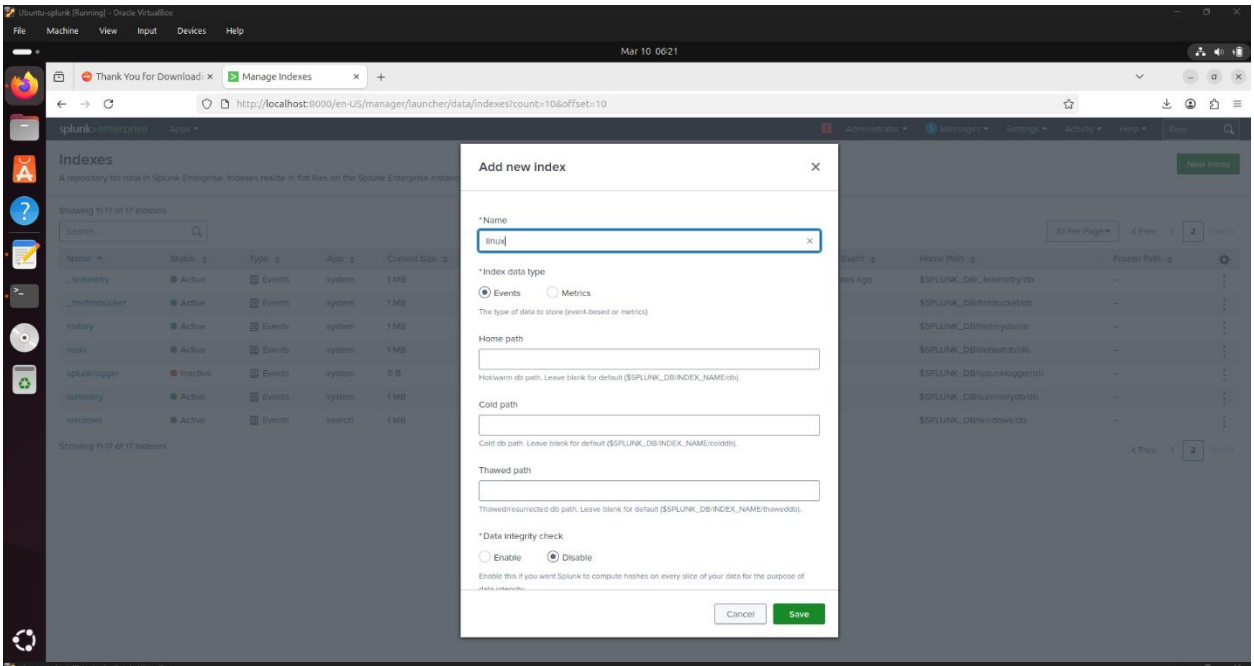
Configured Splunk receiving port on port 9997

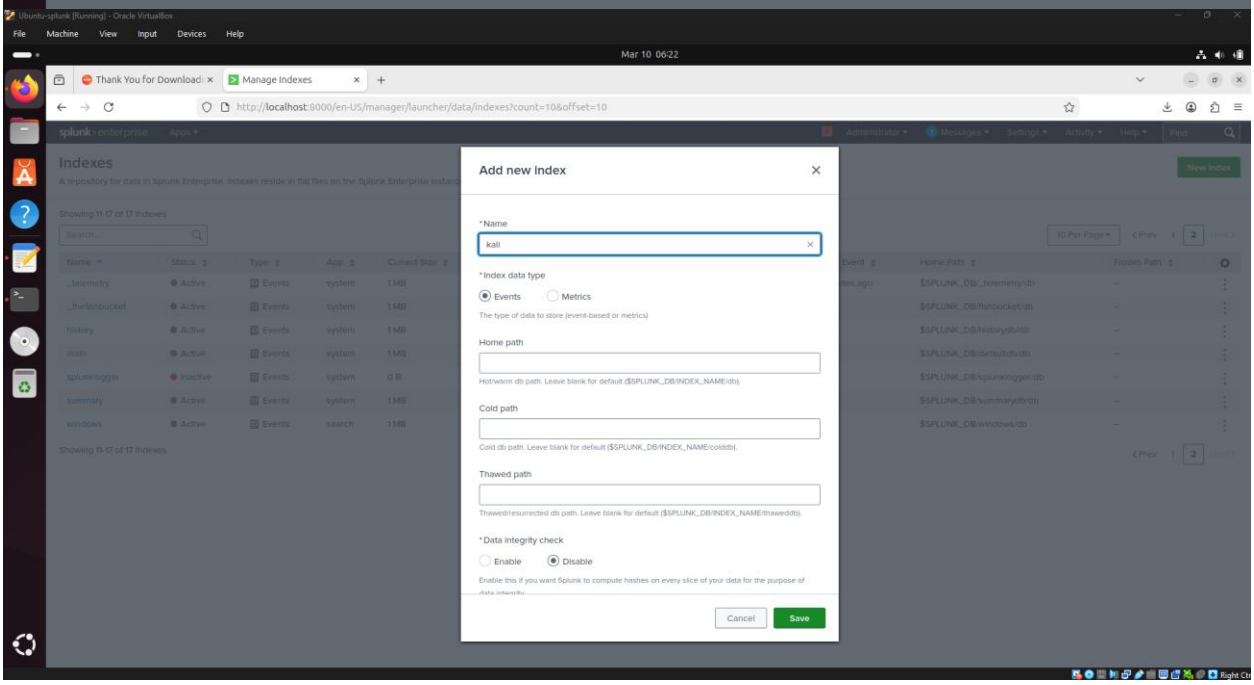


Splunk Index Creation

Creating indexes for the 3 other machines, now I do want to point out that I make a mistake here that will come up later when I setup my splunk forwarder on my ubuntu machine, I write about it more there.



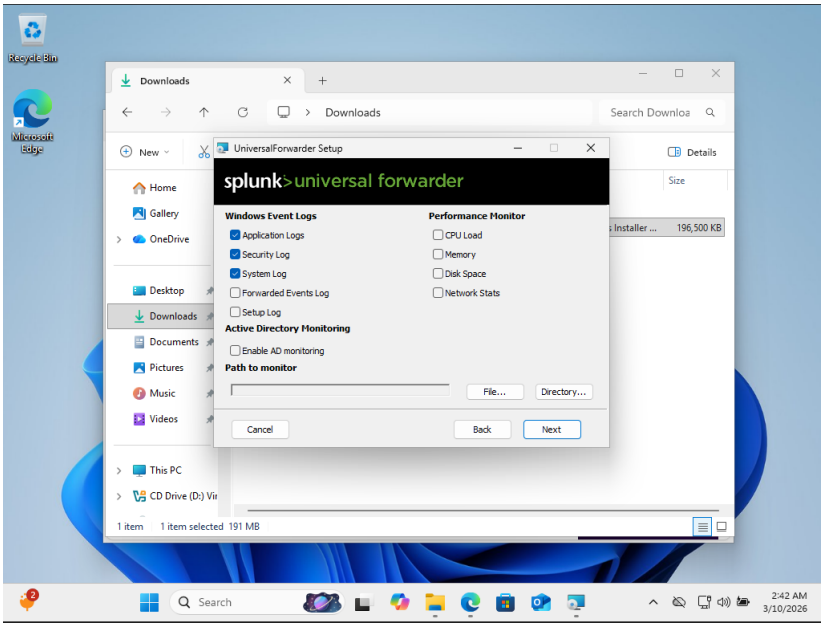

Ubuntu-splunk [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 10 06:21
Thank You for Downloadi
Manage Indexes
http://localhost:8000/en-US/manager/launcher/data/indexes?count=10&offset=10
Indexes
Add new index
*Name
linux
*Index data type
Events
Metrics
The type of data to store (event-based or metrics)
Home path
Hot/warm db path. Leave blank for default ($SPLUNK_DB/INDEX_NAME/db).
Cold path
Cold db path. Leave blank for default ($SPLUNK_DB/INDEX_NAME/colddb).
Thawed path
Thawed/resurrected db path. Leave blank for default ($SPLUNK_DB/INDEX_NAME/thaweddb).
*Data integrity check
Enable
Disable
Cancel
Save


Ubuntu-splunk [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mar 10 06:22
Thank You for Downloadi
Manage Indexes
http://localhost:8000/en-US/manager/launcher/data/indexes?count=10&offset=10
Indexes
Add new index
*Name
kali
*Index data type
Events
Metrics
The type of data to store (event-based or metrics)
Home path
Hot/warm db path. Leave blank for default ($SPLUNK_DB/INDEX_NAME/db).
Cold path
Cold db path. Leave blank for default ($SPLUNK_DB/INDEX_NAME/colddb).
Thawed path
Thawed/resurrected db path. Leave blank for default ($SPLUNK_DB/INDEX_NAME/thaweddb).
*Data integrity check
Enable
Disable
Cancel
Save

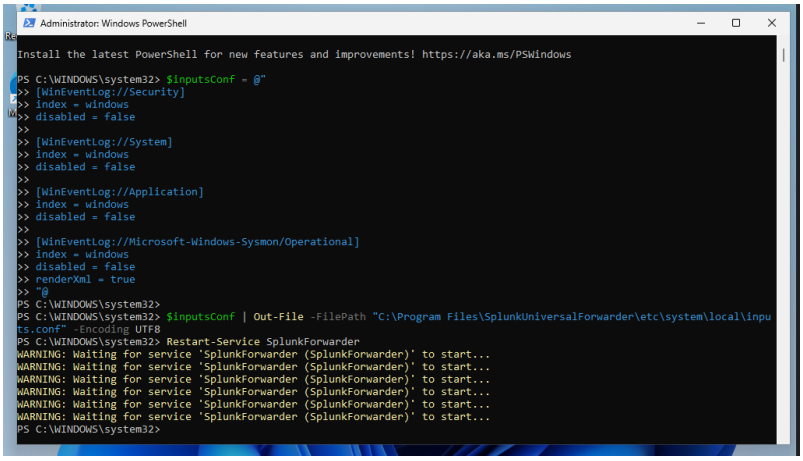
Splunk Forwarder Setup

Windows Setup

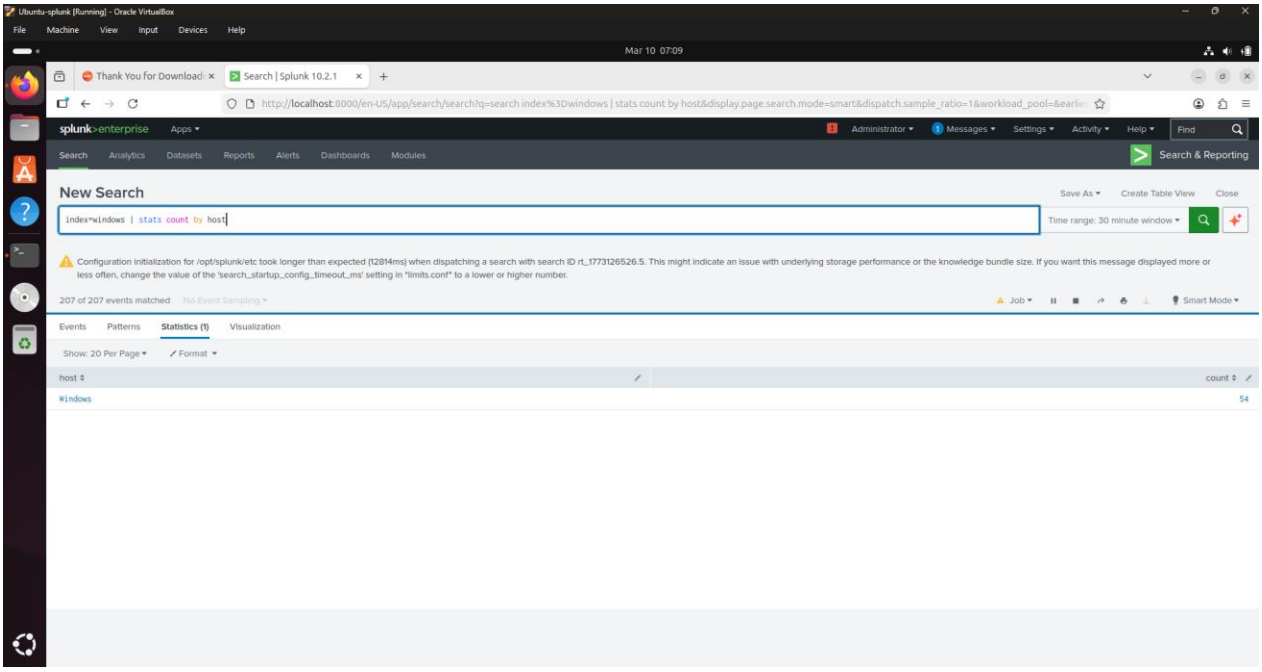
Splunk Forwarder setup on windows VM



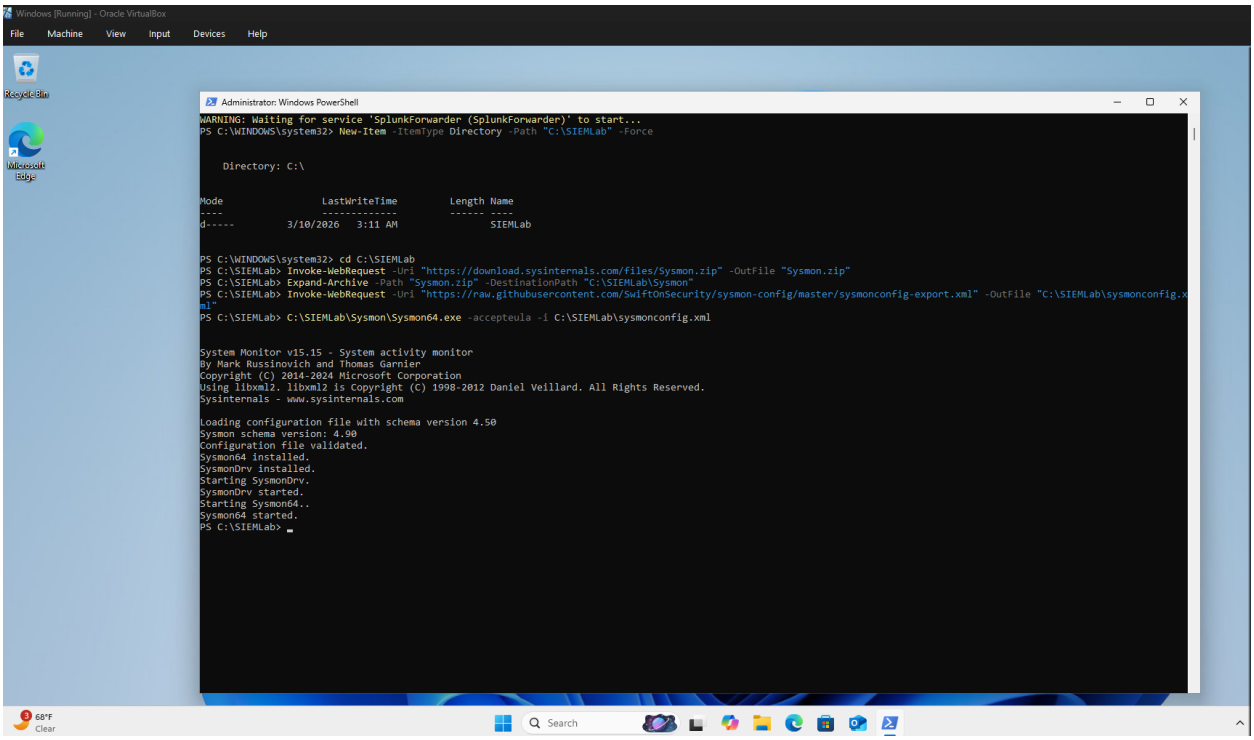
Writing the inputs config



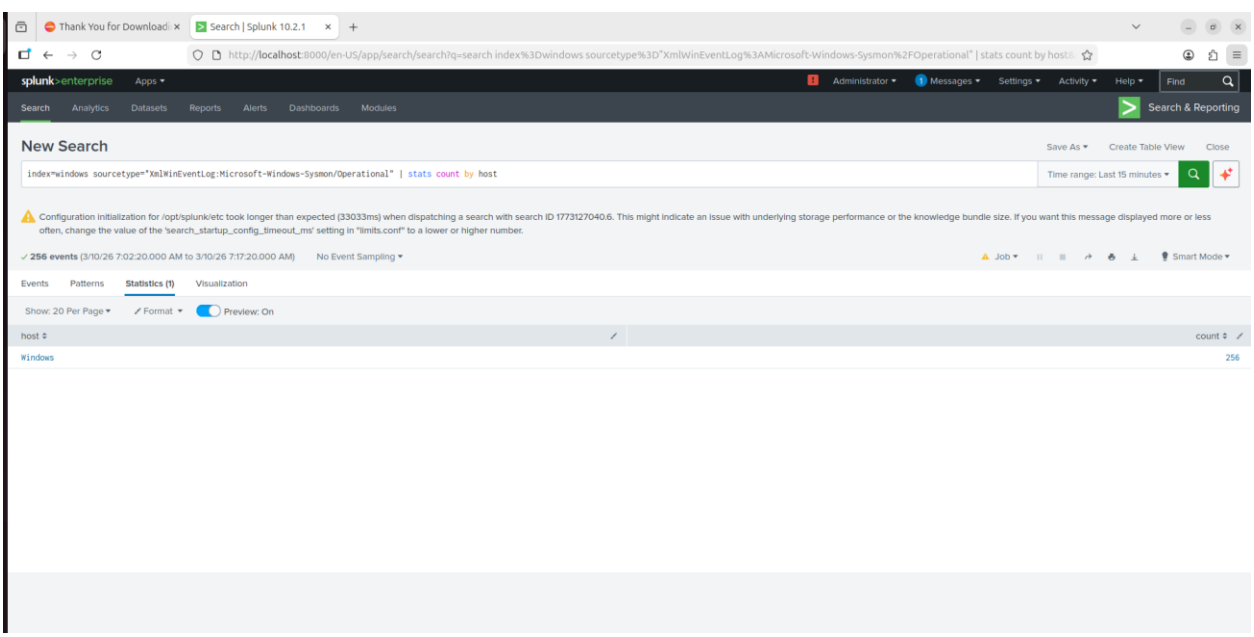
Windows logs are now appearing in Splunk!



I downloaded Sysmon and started it on my windows machine

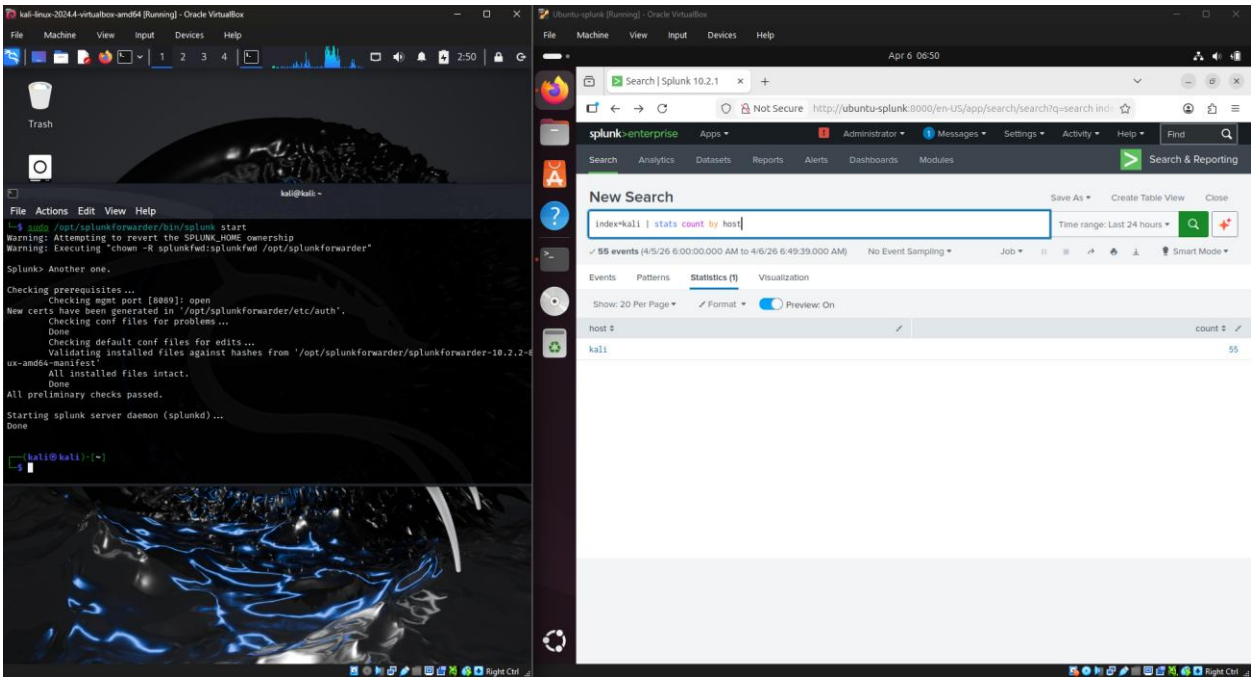


Sysmon logs are flowing to splunk



Kali Setup

Installed Splunk Forwarder on Kali and logs are flowing to the ubuntu server



The inputs.conf file for kali is this:

[monitor:///var/log/auth.log]

index = kali

sourcetype = linux_secure

disabled = false

[monitor:///var/log/syslog]

index = kali

sourcetype = syslog

disabled = false

[monitor:///var/log/audit/audit.log]

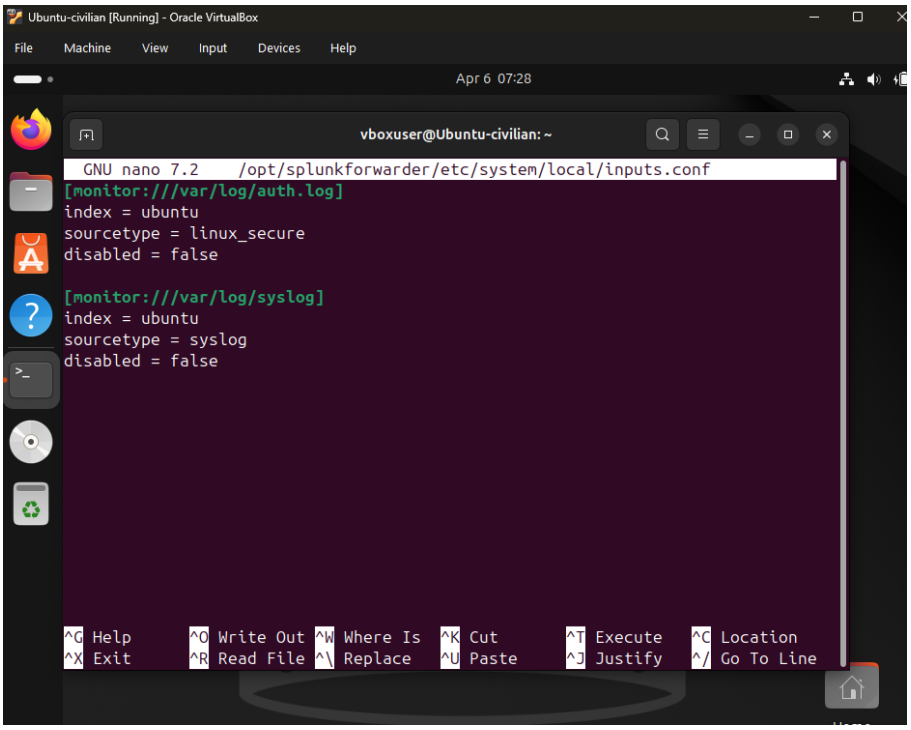
index = kali

sourcetype = linux_audit

disabled = false

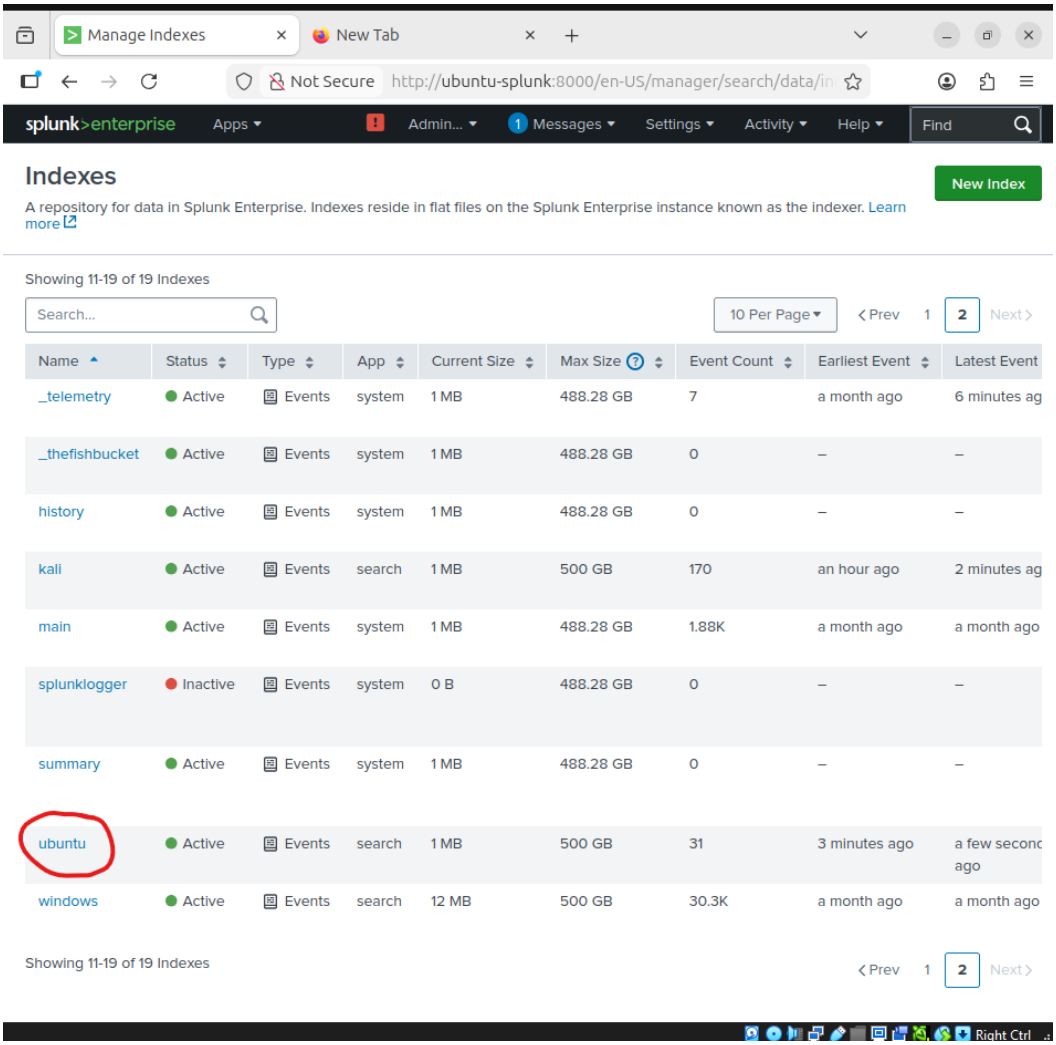
Ubuntu Setup

Installed Splunk forwarder on civilian Ubuntu VM and setup inputs.conf file

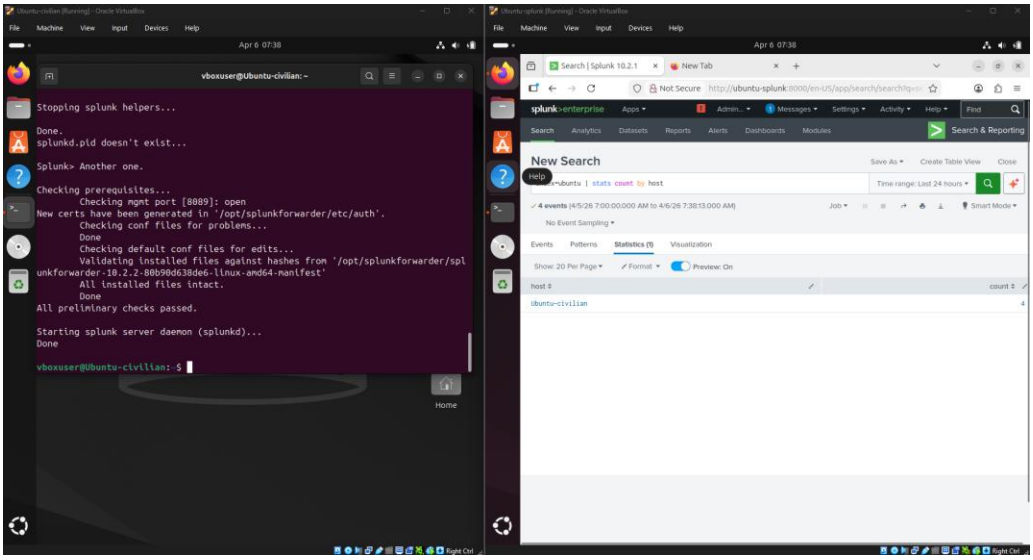


A small problem came up at this step, when I went to check if I could find the ubuntu logs in my splunk server **THERE WERE NO LOGS COMING IN**.

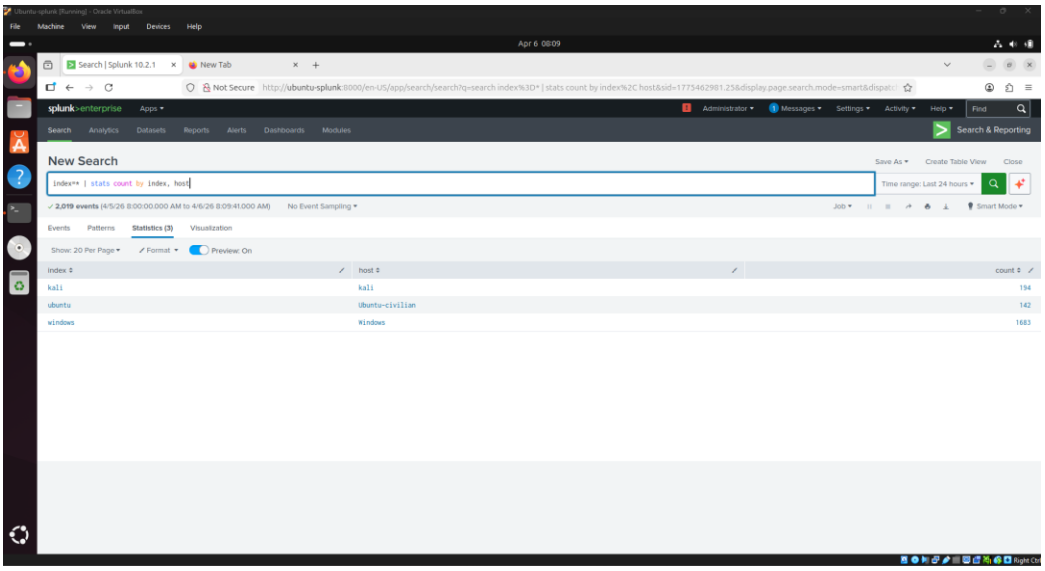
I did accidentally make a mistake here, I forgot that I named the index in my Splunk server as “linux” instead of ubuntu, which ended up causing confusion and panic when I saw no logs were coming in. To fix this I went into my splunk index tab and deleted the “linux” index and instead made a new index named “ubuntu”. Naming the index linux wasn’t very smart anyways as I have both a kali linux and ubuntu linux VM in this lab so naming it ubuntu is clearer.



After the index correction, I go back and check for logs again and wonderfully, they're now coming in, SUCCESS!



All 3 Machines are now setup and properly sending logs to my Splunk server!

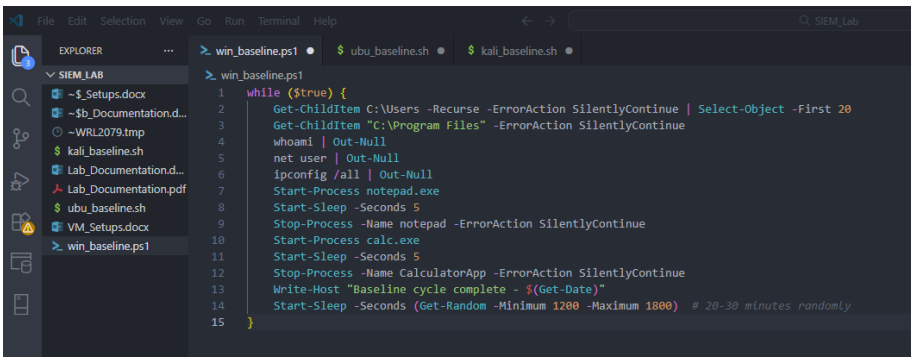


Auto Activity Script Making

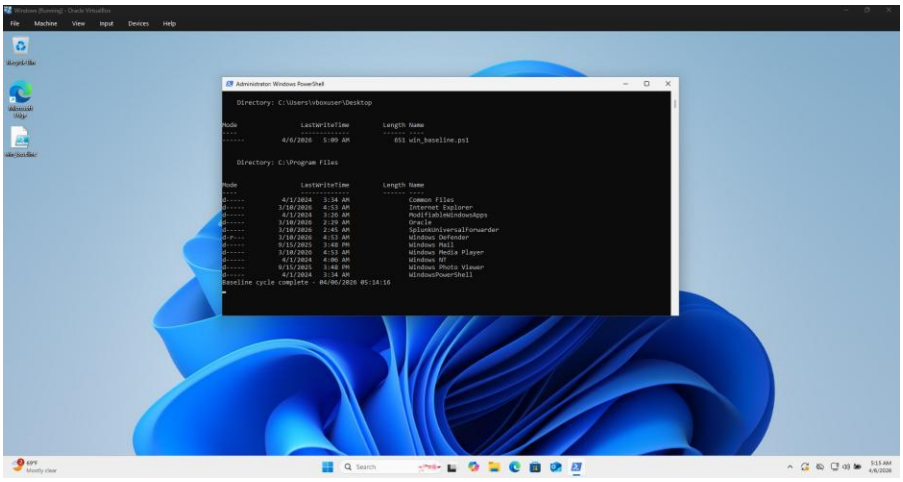
Next I'm going to make shell scripts for each machine to automate mundane baseline activity to automate activity making for my splunk server to receive.

Windows Script

- Named win_baseline.ps1
- Recursively browses the Users and Program Files directories simulating normal file system navigation
- Runs whoami and net user to simulate standard user and account enumeration activity
- Runs ipconfig to simulate network configuration checks
- Launches and closes Notepad and Calculator to generate process creation and termination events in Sysmon
- Sleeps a randomized interval between cycles to simulate organic user behavior. The interval is set to between every 20 to 30 minutes

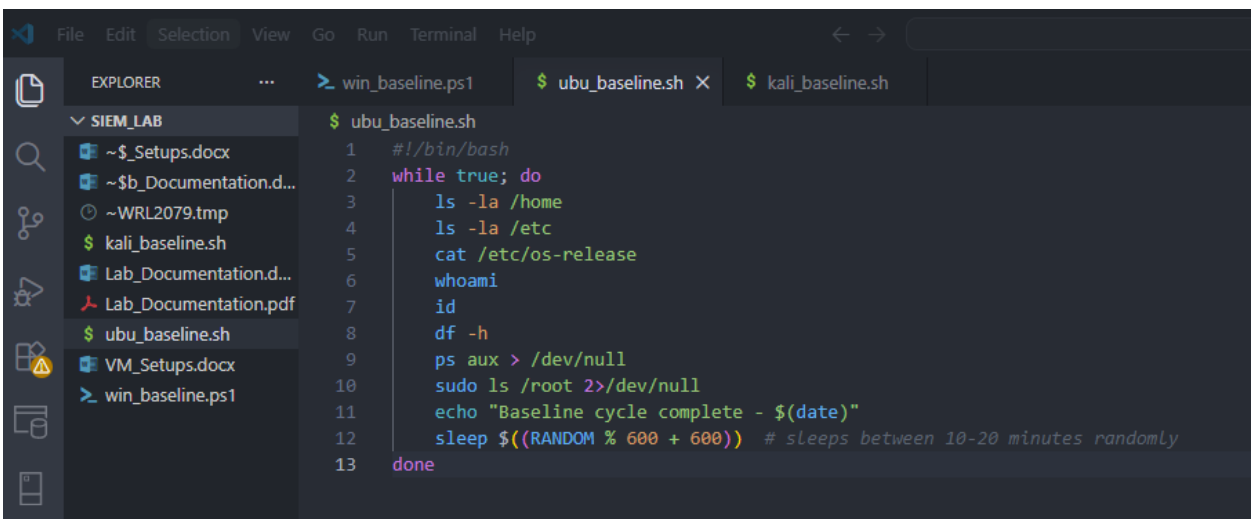


Windows script Running in VM

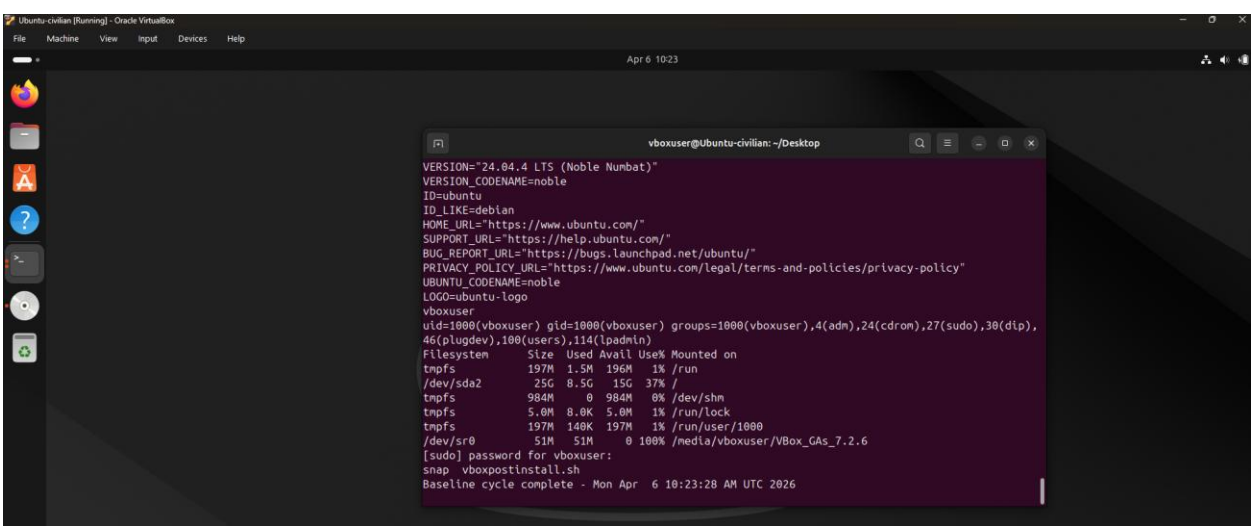


Ubuntu script

- named ubu_baseline.sh
- Browses common system directories (/home, /etc) to simulate routine file navigation
- Runs whoami, id, and uname to simulate normal user session activity
- Checks disk usage with df -h and running processes with ps aux to simulate system monitoring
- Attempts sudo ls /root to generate sudo authentication events in auth.log
- Sleeps a randomized interval between cycles to simulate organic user behavior. The interval is set to between every 10 to 20 minutes



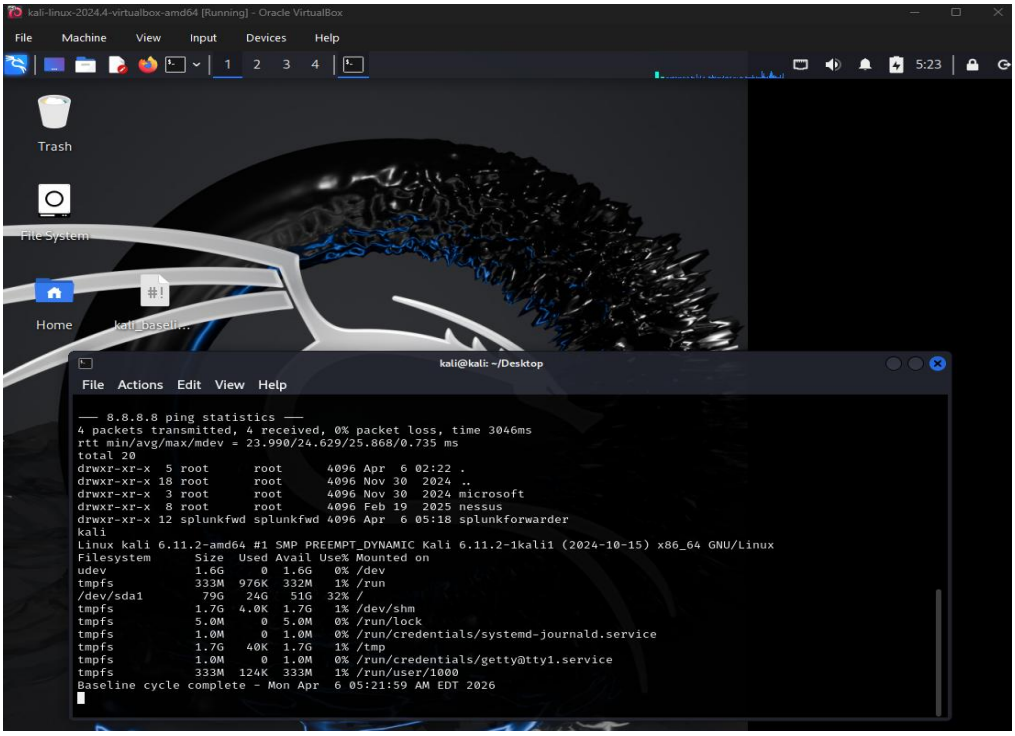
Ubuntu Script Running

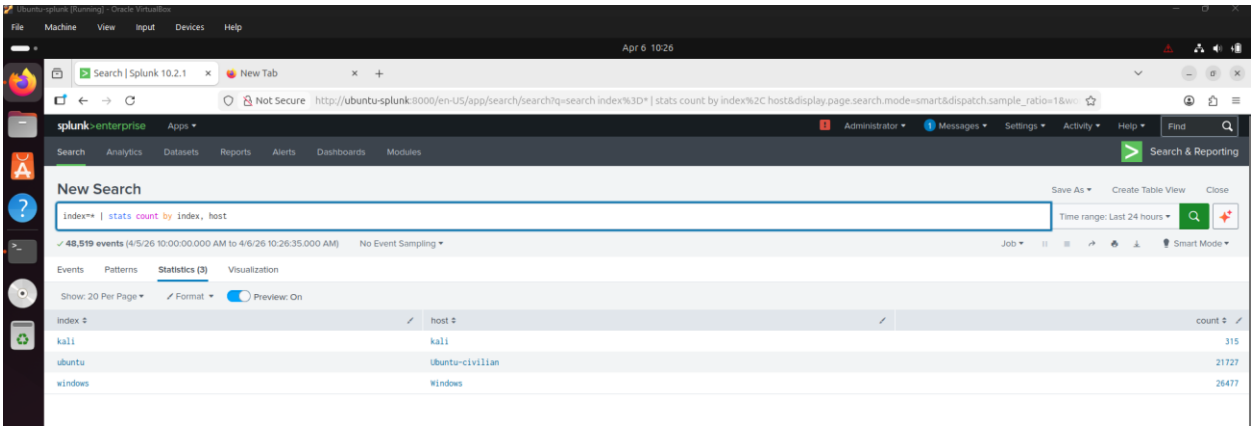


Kali script

- named kali_baseline.sh
- Runs ifconfig to simulate normal network interface checks
- Pings an external address to generate outbound ICMP traffic unrelated to attacks
- Browses tool directories to simulate normal researcher/analyst tool usage
- Checks system info with uname and disk usage with df -h
- Sleeps a randomized interval between cycles to keep activity pattern organic. The interval is between every 10 to 20 minutes

Kali Script Running



Activity Result

Splunk Server picking up activity from all 3 machines